

Problem Set 3

Due on Thursday August 12 at 11:59 PM (if you want feedback)

1. Find all values of $p \in \mathbb{R}$ (if any exist) for which the improper integral

$$\int_e^\infty \frac{1}{(\ln x)^p} dx$$

converges.

2. Determine whether or not the improper integral

$$\int_e^\infty \frac{1}{(\ln x)^{\ln x}} dx$$

converges.

3. Let $\vec{x}, \vec{y}, \vec{v}, \vec{w} \in \mathbb{R}^n$ be given.

- (a) Show that if $\vec{x} \cdot \vec{y} = 0$, then

$$|\vec{x} - \vec{y}| = |\vec{x} + \vec{y}|.$$

- (b) Given that $|\vec{v}| = 1$ and $|\vec{v} - \vec{w}| = |\vec{w}|$, determine $\vec{v} \cdot \vec{w}$ (if possible).

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4. Consider the function $\vec{F} : \mathbb{R} \rightarrow \mathbb{R}^3$ defined by

$$\vec{F}(t) = (e^t \cos t, e^t \sin t, e^t) \text{ for all } t \in \mathbb{R}.$$

Compute

$$\int_0^\pi |\vec{F}'(t)| dt.$$

5. Find the indicated partial derivatives.

(a) $\frac{\partial}{\partial x} \left(e^{-x^4} y^2 \ln(1 + y^2) \right)$

(b) $f_{,2}(a, b, c)$ for $f(x, y, z) = (x^2 + y^2 + z^2) \ln(x + y + z)$

(c) $f_{xxxxxyyy}(x, y)$ for $f(x, y) = y^2 e^{-x^3}$

(d) $u_{x_3}(x_1, x_2, x_3)$ for $u(x_1, x_2, x_3) = \sqrt{x_1 x_2^2 x_3^3}$

(e) $\frac{\partial x}{\partial r}$ and $\frac{\partial x}{\partial \theta}$ for $x = r \cos \theta$.

(f) $f_{xx}(x, y)$ and $f_{xy}(x, y)$ for $f(x, y) = \sin(x^2 + 3y) + x^2y^3 + 2y^2$.

6. Consider the function $g : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by

$$g(x_1, x_2, x_3) = x_1^3x_2 + 2x_1x_3^2 + e^{2x_2}.$$

(a) Find $\vec{\nabla}g(x_1, x_2, x_3)$.

(a) Find the linear approximation for g about the point $\vec{x}^* = (1, 0, 2)$.

(b) Use the linear approximation from part (a) to find an approximate value for $g(1.01, .02, 1.97)$ and compare this approximate value to the actual value of $g(1.01, .02, 1.97)$.

7. Assume that $u(x, y)$ and $v(x, y)$ have continuous second-order partial derivatives and that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Show that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0.$$

8. Consider the function f defined by

$$f(x, y) = 3x^2 - 3xy^2 + y^3 + 3y^2.$$

Find $H_f(-1, 1)$, the Hessian matrix at $(-1, 1)$.

9. Let m be a given positive integer and assume that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ has continuous first-order partial derivatives and satisfies

$$f(t\vec{x}) = t^m f(\vec{x}) \quad \text{for all } \vec{x} \in \mathbb{R}^n, t \in \mathbb{R}.$$

Show that

$$\vec{x} \cdot \vec{\nabla} f(\vec{x}) = m f(\vec{x}) \quad \text{for all } \vec{x} \in \mathbb{R}^n.$$

10. Assume that $f : \mathbb{R} \rightarrow \mathbb{R}$ is twice continuously differentiable and put $w = f(u)$, and $u = x^2 + y^2$. Find an expression for

$$\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}$$

in terms of u , $f'(u)$ and $f''(u)$.